In this section, we briefly discuss the high-level features of a modern search engine. Then the following section introduces a couple of search engines available in the market, including Elasticsearch.

1.3.1 Functionality

Modern search engines were developed to satisfy full-text search requirements while also providing other advanced functions. They are designed to provide fast and relevant search results to users by indexing and searching large volumes of text data (going forward, we will drop the word *modern* when mentioning search engines).

Search engines can quickly index large amounts of text data and make it searchable. This process typically involves breaking the text data into tokens and building an inverted index, which maps each token to the documents that contain it.

Search engines are also expected to perform advanced text analysis and processing, such as synonyms, stemming, stop words, and other natural language processing techniques, to extract meaning from text and improve search results. They can process user queries and rank search results based on various factors such as relevance and popularity. They can also handle high query loads and large amounts of data and can scale horizontally by adding more nodes to a cluster.

Finally, search engines provide advanced analytics capabilities, looking at the data to provide summaries, conclusions, and intelligence for businesses. They also support rich visualizations, near-real-time search, performance monitoring, and machine learning-based insights.

1.3.2 Popular search engines

While a handful of search engines are available in the market, I'll mention just three of them, all of which are built on top of Apache Lucene. The following sections look at Elasticsearch, Solr, and OpenSearch.

ELASTICSEARCH

Shay Banon, founder of Elastic, developed a search product called Compass in early 2000. It was based on an open source search engine library called Apache Lucene (https://lucene.apache.org). Lucene is Doug Cutting's full-text search library, written in Java. Because it's a library, we must import it and integrate it with an application using its APIs. Compass and other search engines use Lucene to provide a generalized search engine service so we don't have to integrate Lucene from scratch into applications. Shay eventually decided to abandon Compass and focus on Elasticsearch because it had more potential.

APACHE SOLR

Apache Solr is an open source search engine that was built on Apache Lucene in 2004. Solr is a strong competitor to Elasticsearch and has a thriving user community, and it is closer to open source than Elasticsearch (Elastic moved from Apache to Elastic License and Server Side Public License ([SSPL] in early 2021). Both Solr and Elasticsearch excel at full-text searching; however, Elasticsearch may have an edge when it comes to analytics.